

UNIT 4



DECIMALS



Decimal

Decimal is a number that contains fractional part. The number of digits at the right-hand side of a decimal point determines the number of decimal places.

Consider the place value of the position of each digit in 765.984. The decimal point separates the whole number part from the fractional part.

Hundreds	Tens	Ones	Decimal	Tenths	Hundredths	Thousandths
7	6	5	.	9	8	4

ADDITION

Example 1.

Add $20.25 + 7.52$

	Tens	Ones	Decimal	Tenths	Hundredths
	2	0	.	2	5
+	0	7	.	5	2
	2	7	.	7	7

Write zero (0) where there is no digit like in the given example.

EXERCISE 4.1

Q1. Add the following.

1) $8.24 + 3.51 =$

14) $0.02 + 0.09 =$

2) $4.67 + 7.2 =$

15) $0.07 + 0.07 =$

3) $2.68 + 3.64 =$

16) $0.09 + 0.08 =$

4) $68.7 + 53.8 =$

5) $6.96 + 0.42 =$

6) $2.88 + 7.45 =$

7) $0.02 + 0.05 =$

8) $0.03 + 0.01 =$

9) $0.06 + 0.02 =$

10) $0.02 + 0.07 =$

11) $0.05 + 0.05 =$

12) $0.08 + 0.03 =$

13) $0.04 + 0.06 =$

SUBTRACTION

Example 1.

Subtract 17.5 from 58.75.

	Tens	Ones	Decimal	Tenths	Hundredths
	5	8	.	7	5
-	1	7	.	5	0
	4	1	.	2	5

Write zero (0) where there is no digit like in the given example.

EXERCISE 4.2

Q1. Subtract the following.

$$\begin{array}{r} 8.5 \\ - 2.1 \\ \hline \end{array}$$

$$\begin{array}{r} 6.4 \\ - 3.2 \\ \hline \end{array}$$

$$\begin{array}{r} 7.3 \\ - 2.3 \\ \hline \end{array}$$

$$\begin{array}{r} 5.7 \\ - 4.1 \\ \hline \end{array}$$

$$\begin{array}{r} 9.2 \\ - 5.1 \\ \hline \end{array}$$

$$\begin{array}{r} 4.6 \\ - 6.1 \\ \hline \end{array}$$

$$\begin{array}{r} 7.8 \\ - 7.1 \\ \hline \end{array}$$

$$\begin{array}{r} 5.6 \\ - 2.2 \\ \hline \end{array}$$

$$\begin{array}{r} 6.6 \\ - 5.4 \\ \hline \end{array}$$

$$\begin{array}{r} 7.1 \\ - 4.4 \\ \hline \end{array}$$

$$\begin{array}{r} 8.9 \\ - 5.5 \\ \hline \end{array}$$

$$\begin{array}{r} 6.6 \\ - 5.4 \\ \hline \end{array}$$

Q1. Subtract the following.

$$\begin{array}{r} 24.95 \\ - 11.83 \\ \hline \end{array}$$

$$\begin{array}{r} 37.79 \\ - 13.54 \\ \hline \end{array}$$

$$\begin{array}{r} 19.68 \\ - 3.43 \\ \hline \end{array}$$

$$\begin{array}{r} 17.67 \\ - 4.52 \\ \hline \end{array}$$

$$\begin{array}{r} 46.19 \\ - 14.55 \\ \hline \end{array}$$

$$\begin{array}{r} 27.87 \\ - 12.62 \\ \hline \end{array}$$

$$\begin{array}{r} 69.93 \\ - 62.57 \\ \hline \end{array}$$

$$\begin{array}{r} 48.29 \\ - 21.65 \\ \hline \end{array}$$

MULTIPLICATION

Multiply decimals by 10, 100, 1000

Example 1.

Find the product.

a) 2.23×10

b) 2.23×100

c) 2.23×1000

Solution:

a) $2.23 \times 10 = 2.230 = 22.30 = \mathbf{22.3}$

What happen to value of 2.23

Multiply a decimal by 10, then the value of the decimals increases ten times.

b) $2.23 \times 100 = 2.2300 = 223.00 = \mathbf{223}$

Multiply a decimal by 100 then the value of the decimals increases hundred times.

c) $2.23 \times 1000 = 2.23000 = 2230.00 = \mathbf{2230}$

Multiply a decimal by 1000, then the value of the decimals increases one thousand times.

Multiply a decimal with a whole number.

Example 1.

Find the product.

a) 0.231×2

b) 4.4×4

c) 2.3×24

Solution:

a) 0.231×2

$$\begin{array}{r} 0.231 \\ \times 2 \\ \hline 0.462 \\ \hline \end{array}$$

b) 4.4×4

$$\begin{array}{r} 4.4 \\ \times 4 \\ \hline 17.6 \\ \hline \end{array}$$

c) 2.3×24

$$\begin{array}{r} 2.3 \\ \times 24 \\ \hline 92 \\ 460 \\ \hline 55.2 \end{array}$$

Multiply a decimal by tenths and hundredths.

One-Tenth = 0.1 and One-Hundredth = 0.01

Example 1.

Find the product.

a) 2.23×0.1

b) 2.23×0.01

c) 2.23×0.5

d) 2.23×0.07

Solution:

a)

$$\begin{array}{r} 2.23 \\ \times 0.1 \\ \hline 0.223 \\ \hline \end{array}$$

b)

$$\begin{array}{r} 2.23 \\ \times 0.01 \\ \hline 0.0223 \\ \hline \end{array}$$

c)

$$\begin{array}{r} 2.23 \\ \times 0.5 \\ \hline 1.115 \\ \hline \end{array}$$

d)

$$\begin{array}{r} 2.23 \\ \times 0.07 \\ \hline 0.1561 \\ \hline \end{array}$$

Multiply a decimal by a decimal (with three decimal places).

Example 1.

Find the product of: 0.25×0.056

Solution:

$0.25 \times 0.056 =$

$$\begin{array}{r}
 0.056 \longrightarrow 3 \text{ decimal places} \\
 \times 0.25 \longrightarrow 2 \text{ decimal places} \\
 \hline
 280 \\
 112 \times \\
 \hline
 0.014
 \end{array}$$

Answer: 0.01400

EXERCISE 4.3

Q1. Find the product.

$ \begin{array}{r} 13.4 \\ \times 3 \\ \hline \end{array} $	$ \begin{array}{r} 12.1 \\ \times 5 \\ \hline \end{array} $	$ \begin{array}{r} 24.5 \\ \times 2 \\ \hline \end{array} $
$ \begin{array}{r} 12.6 \\ \times 3 \\ \hline \end{array} $	$ \begin{array}{r} 10.1 \\ \times 4 \\ \hline \end{array} $	$ \begin{array}{r} 17.3 \\ \times 3 \\ \hline \end{array} $
$ \begin{array}{r} 27.3 \\ \times 2 \\ \hline \end{array} $	$ \begin{array}{r} 21.1 \\ \times 5 \\ \hline \end{array} $	

Q2. Multiply the following.

1) $0.6 \times 3 =$

2) $0.2 \times 9 =$

3) $3 \times 0.7 =$

4) $4 \times 0.8 =$

5) $7 \times 0.6 =$

Q3. Multiply the following.

1) $35.3 \times 10 =$

353

2) $7.23 \times 100 =$

3) $4.37 \times 100 =$

4) $4.27 \times 100 =$

5) $73.3 \times 1000 =$

6) $18.54 \times 10 =$

4) $56.8 \times 10 =$

8) $6.45 \times 100 =$

5) $1.3 \times 10 =$

10) $4.23 \times 100 =$

Q4. Put the decimal point at the appropriate place in each answer.

$$\begin{array}{r} 46.7 \\ \times 1.2 \\ \hline \end{array}$$

5604

$$\begin{array}{r} 0.754 \\ \times 1.2 \\ \hline \end{array}$$

09048

$$\begin{array}{r} 3.57 \\ \times 1.2 \\ \hline \end{array}$$

4284

$$\begin{array}{r} 68.2 \\ \times 1.2 \\ \hline \end{array}$$

8184

$$\begin{array}{r} 0.402 \\ \times 1.2 \\ \hline \end{array}$$

04824

$$\begin{array}{r} 14.839 \\ \times 1.2 \\ \hline \end{array}$$

178068

$$\begin{array}{r} 361.4 \\ \times 3.54 \\ \hline \end{array}$$

1279356

$$\begin{array}{r} 219.241 \\ \times 2.5 \\ \hline \end{array}$$

5481025

$$\begin{array}{r} 402.58 \\ \times 2.3 \\ \hline \end{array}$$

925934

DIVISION

Divide decimals by 10, 100, 1000

Rules:

1. Divide a decimals by 10, then the value of the decimals decreases ten times.

$$\begin{array}{c} \uparrow \\ 15.34 \end{array} \div 10 = 1.534$$

(What happened to value of 15.35?)

2. Divide a decimal by 100 then the value of the decimals decreases hundred times.

$$15.34 \div 100 = 0.1534$$

3. Divide a decimals by 1000, then the value of the decimals decreases one thousand times.

$$15.34 \div 1000 = 0.01534$$

Example 1.

Solve by using the above stated rules:

a) $0.0175 \div 10$

b) $0.0175 \div 100$

c) $0.0175 \div 1000$

Solution:

We can observe that the value of decimals

(a) $0.0175 \div 10 = 0.00175$ (decreases ten times)

(b) $0.0175 \div 100 = 0.000175$ (decreases hundred times)

(c) $0.0175 \div 1000 = 0.0000175$ (decreases thousand times)

Divide a decimal with a whole number.

Example 1. Solve: $0.9 \div 3$

Solution $0.9 \div 3 = 9 \text{ Tenths} \div 3$

$$= (9 \div 3) \text{ Tenths} = 3 \text{ Tenths} \\ = 0.3$$

$$\begin{array}{r} 3 \overline{) 0.9} \quad 0.3 \\ \underline{- 9} \\ 0 \end{array}$$

Example 2. Solve: $0.84 \div 4$

Solution:

$$\begin{aligned} 0.84 \div 4 &= (0.80 + 0.04) \div 4 \\ &= (8 \text{ tenths} \div 4) + (4 \text{ hundredths} \div 4) \\ &= 2 \text{ tenths} + 1 \text{ hundredth} \\ &= 20 \text{ hundredth} + 1 \text{ hundredth} = 21 \text{ hundredth} \\ &= 0.21 \end{aligned}$$

$$\begin{array}{r} 4 \overline{) 0.84} \quad \mathbf{0.21} \\ \underline{-8} \\ 4 \\ \underline{-4} \\ 0 \end{array}$$

Divide a decimal by a decimal.

Example:

Solve the following decimals using direct division method by moving decimal positions:

a) $0.55 \div 0.05$

b) $0.125 \div 0.5$

c) $2.25 \div 0.3$

Solution:

(a) $0.55 \div 0.05 = 55 \div 5 = \mathbf{11}$

(Both the decimals are with two decimal places).

So we get whole number.

$$\begin{array}{r} 11 \\ \cancel{0.55} \\ \cancel{0.05} \\ 1 \end{array} = 11$$

$$\begin{array}{r} 11 \\ 5 \overline{) 55} \\ \underline{-55} \\ 0 \end{array}$$

(b) $0.125 \div 0.5 = 1.25 \div 5$

3 decimal places – 1 decimal place

= 2 decimal places

= **0.25**

$$\begin{array}{r} 25 \\ \cancel{0.125} \\ \cancel{0.5} \\ 1 \end{array} = 0.25$$

$$\begin{array}{r} \mathbf{0.25} \\ 5 \overline{) 1.25} \\ \underline{-10} \\ 25 \\ \underline{-25} \\ 0 \end{array}$$

(b) $2.25 \div 0.3 = 22.5 \div 3 = 7.5$

4 decimal places – 1 decimal place
= 3 decimal places

$$\frac{2.25}{0.3} = \frac{22.5}{3} \\ = 7.5$$

EXERCISE 4.4

Q1. Divide the following.

1) $7.1 \div 10 =$

5) $55.2 \div 10 =$

2) $7.7 \div 10 =$

6) $8.3 \div 10 =$

3) $0.681 \div 10 =$

7) $2.5 \div 10 =$

4) $0.084 \div 10 =$

8) $599.5 \div 10 =$

Q2. Solve the following.

1) $36.4 \div 8 =$

5) $5.026 \div 10 =$

2) $35.50 \div 5 =$

6) $4.26 \div 3 =$

3) $21.39 \div 10 =$

7) $18.24 \div 6 =$

4) $56.10 \div 8 =$

SIMPLIFICATION

Decimal expressions involving brackets

Example 1: Simplify: $24.24 - (5.6 + 20.25 - 4.45)$

Solution: First solve the expressions in the bracket.

$$\begin{aligned} &24.24 - (5.6 + 20.25 - 4.45) \\ &= 24.24 - (25.85 - 4.45) \\ &= 24.24 - 21.40 \\ &= 2.84 \end{aligned}$$

(Operations of addition are performed, then subtraction in operations are performed)

Example 2: Simplify: $52.05 + (28.22 - 22.6) + (13.15 - 6.56)$

Solution: First solve the expressions in the bracket.

$$\begin{aligned} 52.05 + (28.22 - 22.6) + (13.15 - 6.56) &= 52.05 + 5.62 + 6.59 \\ &= 52.05 + 12.21 \\ &= 64.26 \end{aligned}$$

EXERCISE 4.5

Q1. Simplify:

- 1) $5.6 + 7.22 - (2.24 + 4.68)$
- 2) $(20.14 \times 5.6) + 10.9335$
- 3) $4.6 + 6.07 + (23.35 - 8.30 + 8.34)$
- 4) $14.3 - 2.8 + (1.84 + 3.29)$
- 5) $(5.24 + 2.02) - 0.96 - 7.45 + (9.405 - 2.24)$
- 6) $5.6 \times (25.5 - 12.2) + (2.3 + 2.6)$
- 7) $3.45 \times 8.56 - 4.23 + (2.2 - 1.12)$
- 8) $45.234 + (18.024 - 6.66) - (0.457 + 9.945)$

Q1. Simplify:

1) $(7.1)^2 - 6.7 \times 3.1$

2) $(6.3)^2 + 1.9 \times 5.3$

3) $4.2 \times 6.9 - (2.9)^2$

4) $(3.6)^2 + 1.7 \times 5.1$

5) $(9.5)^2 \div (5.6 - 4.6)$

6) $(2.8)^2 \div 1.6 + 8.7$

7) $(8.1)^2 + 6.7 \times 3.7$

8) $7.5 \times 4.6 - (2.8)^2$

WORD PROBLEMS

- 1 Sabir bought two chickens of weight 1.450 kg and 1.685 kg respectively. Find the total weight of both chickens?
- 2 In a general store 200.750 kg of flour and 98.500 kg of sugar were sold. How much more flour was sold than sugar?
- 3 Sara bought a necklace of gold of mass 2.565g and bangles of mass 8.875g. What is the total mass of these two items?
- 4 Aftab saved Rs 3,206.75 from monthly salary of Rs 35,200. Find his monthly expenditure?

ROUND OFF

Rule:

In order to round off a decimal nearest to the whole number.

(i) Check the first decimal place or tenths digit to see if it is less or greater than 5.

(ii) If it is less than 5, it will remain unchanged.

(iii) If it is equal to or greater than 5, increase the whole number by 1.

EXERCISE 4.6

Q1. Round off the following decimals as whole numbers.

1) $2.3 =$

4) $7.7 =$

2) $5.6 =$

5) $8.3 =$

3) $6.6 =$

6) $50.2 =$

Q2. Round off the following decimals up to one decimal and two decimal places.

1) $32.38 =$

4) $76.798 =$

2) $25.156 =$

5) $25.056 =$

3) $6.171 =$

6) $6.775 =$

Q3. Round of these numbers to the nearest tenth.

1) $3.47 =$

7) $1.16 =$

2) $2.41 =$

8) $0.84 =$

3) $7.63 =$

9) $7.72 =$

4) $6.18 =$

10) $0.56 =$

5) $5.75 =$

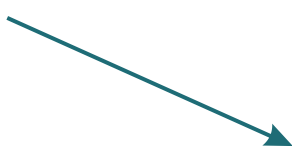
11) $6.02 =$

6) $4.9 =$

12) $3.39 =$

Q4. Match each number to its nearest hundredth.

0.526



0.523

0.544

0.585

0.562

0.51

0.52

0.53

0.54

0.55

0.56

0.57

0.58

0.59

0.60

0.507

0.545

0.566

0.581

0.598

Fraction to decimal or vice versa

Decimals and fractions are two ways to represent numbers, but they have distinct difference. Fraction represent a part of a whole while decimal represent a number in term of tenths, hundredths, thousands, etc.

Q. Convert the following fraction to decimals:

$\frac{15}{100}$		$\frac{28}{100}$	
$\frac{42}{100}$		$\frac{17}{100}$	
$\frac{78}{100}$		$\frac{6}{100}$	
$\frac{62}{100}$		$\frac{97}{100}$	
$\frac{89}{100}$		$\frac{37}{100}$	
$\frac{56}{100}$		$\frac{22}{100}$	
$\frac{7}{100}$		$\frac{53}{100}$	

Q. Convert the following decimal to fractions:

0.1	_____	0.2	_____
0.11	_____	0.3	_____
0.5	_____	0.6	_____

0.7	_____	0.8	_____
0.9	_____	0.12	_____
0.14	_____	0.17	_____

PERCENTAGE

The word "Percent" is formed of two words, "Per" and "Cent".
"Per" means out of and "Cent" means hundred. Therefore, the PERCENT is used for one hundred and represented by %.

1% makes $\frac{1}{100}$ means 1 out of 100 = 0.01

10% makes $\frac{10}{100}$ means 10 out of 100 = 0.1

and 100% makes $\frac{100}{100}$ means 100 out of 100 = 1

Thus, percentage is a special kind of fraction.

EXERCISE 4.7

Q1. Express the following decimals as percentages.

1) $0.56 =$

2) $0.20 =$

3) $0.09 =$

4) $0.77 =$

5) $0.35 =$

6) $0.6 =$

7) $0.28 =$

8) $0.12 =$

9) $0.90 =$

10) $0.16 =$

11) $0.81 =$

12) $0.07 =$